Sameer Khan

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Education

Bachelor of Technology – Instrumentation & Control Engineering **August 2023 – Present**
National Institute of Technology, Jalandhar

Experience

Indian Oil Corporation Limited (IOCL) – Bottling Plant Intern

- Gained practical exposure to automation systems used in LPG bottling plant operations
- Observed automated processes including filling, sealing, leak detection, and conveyor-based material handling
- Understood the role of sensors, actuators, PLC-based control, and safety interlocks in large-scale industrial plants
- Learned how process flow, safety systems, and automation improve efficiency and reliability in high-capacity plants

Projects

OneVote – College Voting System GitHub
MongoDB, Express.js, React.js, Node.js

- Built a full-stack web application for conducting secure college elections.
- Implemented authentication, role-based authorization, election management, and vote integrity checks.
- Enabled Junior CR elections with 150+ student participants through a digital voting platform.
- Designed an intuitive admin interface for candidate management and automated result declaration.

Customer Order Support Agent GitHub
Python, LangChain, LangGraph, Pydantic

- Built a multi-turn AI agent handling order status, cancellations, returns, and address updates.
- Architected a 4-layer separation of concerns across data access, business rules, tools, and agent orchestration.
- Enforced eligibility and safety rules including status gating, return windows, and data isolation at the code level.
- Gated irreversible actions behind human approval using LangGraph's interrupt-based middleware.

Technical Skills

Programming Languages: JavaScript, C/C++, Python, SQL
Frontend Development: React.js, Next.js, HTML, CSS, Tailwind CSS
Backend Development: Node.js, Express.js, RESTful APIs
Databases: MongoDB, MySQL, SQLite
Developer Tools & Platforms: Git, GitHub, VS Code
Core Engineering: Sensors & Transducers, Control Systems, Basic Signal Processing, Electrical Machines
Embedded & Automation: ESP32, Arduino, PLC (Ladder Logic)
Software & Tools: MATLAB, LTspice, Blynk IoT Platform

Achievements & Leadership

- Solved 400+ algorithmic problems on GFG / LeetCode, with a peak contest rating of 1493 (Top 46.63%).
- Secured Third Place out of 200+ participants in the CYDO Hack CTF (CyberDojo challenge).
- Management Team Member, Cyber Security Club 2024 – Present
- Smart India Hackathon (SIH) 2025 — National-Level Nominee